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Does It Matter? Isometric or Isokinetic Assessment of Quadriceps Strength Symmetry 9 Months After ACLR in Collegiate Athletes

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Background: Greater quadriceps strength symmetry is associated with better outcomes after anterior cruciate ligament reconstruction (ACLR). Isometric and isokinetic assessments of quadriceps strength inform therapeutic exercise prescription and return-to-sport decisions. It is unclear whether isometric and isokinetic measures provide similar information post-ACLR.

Hypothesis: Quadriceps strength symmetry is similar between isometric and isokinetic assessments. Isokinetic and isometric strength symmetries have similar associations to functional knee kinetics and self-reported knee function.

Study Design: Cross-sectional study.

Level of Evidence: Level 3.

Methods: NCAA Division I athletes (N = 35), 8.9 ± 2.5 months post-ACLR completed isometric and isokinetic quadriceps strength assessments, countermovement jumps (CMJs), and treadmill running. Self-reported knee function was assessed using the International Knee Documentation Committee Subjective Knee Form (IKDC). Agreement between isometric and isokinetic strength symmetry was assessed using Bland-Altman analysis, with associations to functional knee kinetics and IKDC assessed using Pearson correlations and linear regressions.

Results: Mean difference in quadriceps strength symmetry between isokinetic and isometric assessments was 1.0% (95% limits of agreement of -25.1% to 23.0%). Functional knee kinetics during running and CMJ were moderately to strongly associated with isometric strength symmetry (r = 0.64-0.80, $P < 0.01$) and moderately associated with isokinetic strength symmetry (r = 0.41-0.58, $P < 0.01$). IKDC scores were weakly to moderately associated with isometric (r = 0.39, P = 0.02) and isokinetic (r = 0.49, $P < 0.01$) strength symmetry.

Conclusion: Isokinetic and isometric assessments of quadriceps strength symmetry in collegiate athletes 9 months post-ACLR demonstrated strong agreement. Quadriceps strength symmetry is associated with functional knee kinetic symmetry post-ACLR.

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The following author declared potential conflicts of interest: B.C.H. has received consulting fees from Biocore, research support from the NFL and GE Healthcare, and has ownership interest in Science of Running Medicine, LLC.

National Institutes of Health provided support (award Nos. TL1TR002375 and T32AG000213).

DOI: 10.1177/19417381241247819

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Clinical Relevance: Considerable individual variation suggests mode of contraction should be consistent throughout postoperative assessment. Isometric strength symmetry may be a better indicator of functional knee kinetic symmetry, while isokinetic strength symmetry may be associated more closely with patient-reported outcomes.

Keywords: ACL; isokinetic; isometric; quadriceps

Quadriceps muscle weakness is universal after anterior cruciate ligament (ACL) injury and surgical reconstruction (ACLR),[31] and can contribute to abnormal lower extremity movement biomechanics,[14,20,28,30,39,44,55] increased re-injury risk,[16] and poor patient-reported outcomes.[18,36] As such, resolving quadriceps weakness is an integral component of postoperative rehabilitation. Current clinical guidelines recommend achieving quadriceps strength symmetry of ≥90% to facilitate a safe and successful return to sport.[15] Thus, accurate and valid assessment of quadriceps strength in athletes post-ACLR is required to guide rehabilitation efforts and improve postoperative outcomes.

Quadriceps strength, as measured via voluntary knee extension torque production, can be quantified accurately with various modes of contraction.[1] In clinical settings, dynamometers are utilized frequently for isokinetic assessment of knee extension torque production after ACLR.[60] Conversely, isometric assessment is typically performed to quantify peak torque at a single, pre-established joint angle and is often used in concert with more in depth evaluations of quadriceps neuromuscular function, such as muscle activation patterns via surface electromyography,[11] voluntary activation testing,[56] and rate of force/torque development.[33] The related ACLR literature is comprised largely of both isokinetic and isometric assessments of quadriceps strength,[8] but the lack of direct comparison between these modes of contraction makes it unclear whether previously reported quadriceps strength symmetry values are influenced strongly by mode of contraction.

In addition, previously published literature has established a relationship between isokinetic knee extension torque and movement mechanics,[21,39] performance tests,[13,18,26,46,61] and patient-reported outcomes after ACLR.[3,18,34,36,61] Similarly, isometric knee extension torque has been associated with post-ACLR movement mechanics,[20,28,30,44,52] performance tests,[9,47] and patient-reported outcomes.[9,41,63] However, it is unclear whether the strength of the relationships between quadriceps performance and functional outcomes after ACLR are dependent on knee extension contraction type.

Thus, the purpose of this study was 3-fold: (1) to determine the relationship between isometric and isokinetic assessment of quadriceps strength in collegiate athletes 9 months post-ACLR; (2) to determine whether isometric or isokinetic quadriceps strength symmetry is associated more closely with functional knee kinetics; and (3) to determine whether isometric or isokinetic quadriceps strength symmetry is associated more closely with patient-reported knee function. We hypothesized that there would be no difference in quadriceps strength symmetries between isometric and isokinetic assessment. Further, we hypothesized that there would be no difference in the strength of association between isokinetic and isometric quadriceps strength symmetry, functional knee kinetics, and self-reported knee function. This investigation will provide valuable information to assist clinicians in selecting and interpreting the results of quadriceps strength evaluations in athletic populations post-ACLR.

METHODS

Participants

Athletes for this study were identified through an analysis of routinely, prospectively collected data from the Badger Athletic Performance database. This database consists of performance and healthcare data from National Collegiate Athletic Association Division I athletes at the University of Wisconsin-Madison. This records review was approved by the University of Wisconsin's Health Sciences Institutional Review Board. An athlete's data were extracted if the subject (1) had undergone an ACLR; (2) had completed running, jumping, isometric, and isokinetic testing on the same date post-ACLR; and (3) did not have a follow-up lower extremity surgery after ACLR but before the test session.

Movement Analyses

Three-dimensional motion analysis was performed while athletes ran and jumped on an instrumented treadmill (Bertec Corp). A total of 42 reflective markers were placed on the body segments of each athlete, of which 23 were located on anatomic landmarks.[17] A static standing posture, with 10 additional reflective markers placed, was recorded to identify joint centers and for model scaling. Marker positions were collected at 200 Hz using an 8-camera passive marker system (Motion Analysis Corp), while ground-reaction forces (GRFs) were synchronously collected at 2000 Hz.

The knee joint kinetic variable of interest, peak knee extensor moment (PKEM), was calculated using biomechanical modeling and analyses as described previously.[17] Kinematic data and GRF data were low-pass filtered using a bidirectional, fourth-order Butterworth filter with a cutoff frequency of 12 Hz. A 14-segment articular linkage that was scaled to the athlete's height, mass, and segment length was utilized to model the body. To calculate joint angles, at each timestep a global optimization routine was used to minimize the weighted sum of squares difference between the measured and the modeled markers. A segment-by-segment inverse dynamics approach was used to calculate joint moments.

Running

For running collections, a standardized protocol was utilized in which each athlete first acclimated to the treadmill and motion

capture setup by walking for 2 minutes.[57] Athletes then transitioned to running and speed was incrementally increased beginning with 10 minutes per mile (1.6 km) in 1 minute per mile intervals until each athlete reached their self-reported maximally comfortable speed. Athletes acclimated to each speed for at least 30 seconds before 15 seconds of kinematic and GRF data were synchronously recorded. A threshold of 50 N for vertical GRF was used to ascertain initial foot contacts and toe-offs, which were used for identifying stride cycles of running.[57] PKEM during stance phase was averaged across 15 strides for the surgical and nonsurgical limb ($PKEM_{RUN}$). A limb symmetry index (LSI) was constructed and defined as follows: (Surgical Limb ÷ Nonsurgical Limb) × 100.

Jumping

After the running assessment, athletes completed countermovement jumps (CMJ). Athletes stood with 1 foot on independent force plates (Bertec Corp) and were instructed to perform a maximal effort CMJ while keeping their hands on hips. Kinematic and GRFs were recorded synchronously for a minimum of 3 successful jumps (hands stayed on hips, landed with 1 foot entirely on each force plate). For the CMJ, the start of the takeoff phase was identified when the vertical velocity of the center of mass exceeded 0 m/s. The takeoff phase ended when the vertical GRF dropped below 50 N, and the landing phase began when the vertical GRF exceeded 50 N. PKEM was identified during the takeoff ($PKEM_{TO}$) and landing ($PKEM_{LAND}$) phases for each trial, and the values from the 3 trials were averaged separately for each limb. As the CMJ is a bilateral task, the LSI was calculated as follows: 1 − (Nonsurgical Limb − Surgical Limb) ÷ (Nonsurgical Limb + Surgical Limb) × 100.[6]

Isometric Strength Assessment

Isometric strength testing was performed after the treadmill running and CMJ trials. Athletes were positioned securely with the knee and hip at 90° and 85° of flexion, respectively, on an isokinetic dynamometer (Biodex System 4, Biodex Medical Systems). The knee adapter pad was reversed from the standard orientation to reduce compliance and was secured to the shank 5 cm proximal to the medial malleolus. The nonsurgical limb was always tested first. Athletes completed progressive warmup repetitions at 50%, 75%, and 90% of maximum effort followed by 2 5-second maximal isometric knee extension contractions. Loud verbal encouragement and visual feedback of the torque signal were provided to facilitate maximal torque production. If >10% difference in peak torque was observed between the 2 trials, subsequent trials were repeated until 2 trials with peak torque within 10% were obtained. The 2 trials with the highest peak torques were subsequently analyzed. Torque was sampled at 2000 Hz, rectified, and lowpass filtered at 60 Hz, and baseline offset was removed. For each limb, the global maximum knee extension torque across 2 trials was identified using a custom Matlab script (Release 2018b, Mathworks, Inc). The isometric peak torque LSI ($ISOM_{LSI}$) was computed as follows: (Surgical Limb ÷ Nonsurgical Limb) × 100.

Isokinetic Strength Assessment

Using the same equipment as the isometric testing, isokinetic strength testing was performed on the nonsurgical limb first, with the knee adapter pad in the manufacturer's designed orientation. Similar to isometric assessment, a progressive warm-up of 50%, 75%, and 90% of maximal effort was performed to familiarize the athlete to the test. Concentric knee extension/flexion isokinetic testing was performed at 60° per second from maximally comfortable knee flexion through maximally comfortable extension range of motion. Athletes were instructed to kick and pull as hard as possible through the entire range of motion for 5 consecutive repetitions. Biodex Advantage Software was utilized for isokinetic assessments. Torque signals were sampled at 100 Hz, and the peak torque was reported as the global maximum of the 5 repetitions. Isokinetic knee extension torque LSIs ($ISOK_{LSI}$) were calculated as follows: (Surgical Limb ÷ Nonsurgical Limb) × 100.

Self-Reported Knee Function

Self-reported knee function was assessed using the International Knee Documentation Committee Subjective Knee Form (IKDC). The IKDC score ranges from 0 to 100, with 100 representing no limitation with daily or sport activities and no knee symptoms.[19]

Statistical Analysis

Multiple statistical approaches were utilized to assess the agreement between isometric and isokinetic quadriceps strength assessments as a result of the inherent limitations of each individual method.[2] Bland-Altman plots of the differences between isometric and isokinetic strength LSIs were constructed to evaluate the 95% limits of agreement.[7] The intraclass correlation coefficient (ICC) was computed using a 2-way mixed model with absolute agreement as a measure of reliability, and the root-mean-square error (RMSE) between LSIs was calculated. The associations between quadriceps strength symmetry, $PKEM_{RUN}$, $PKEM_{TO}$, $PKEM_{LAND}$, and IKDC scores were assessed using Pearson correlations and simple linear regressions.[53] Statistical significance was established a priori at $\alpha \leq 0.05$. Analyses were conducted using the R Statistical language (Version 4.0.5; R Core Team, 2021).[48]

RESULTS

Participant Demographics

In total, 35 athlete records met the inclusion criteria for this study. Athletes participated in basketball (4), spirit squad (3), football (7), softball (2), crew (2), soccer (10), track and field (3), volleyball (3), and wrestling (1). Of the 35 athletes, 8 had undergone previous ACLR(s), of which 6 were secondary (4 contralateral and 2 ipsilateral), 1 tertiary (all ipsilateral), and 1 quaternary (1 contralateral, 2 ipsilateral). Concomitant meniscal procedures included 9 meniscectomies and 7 meniscal repairs. In addition to ACLR, 2 athletes underwent medial patellar femoral ligament reconstruction and 2 had multiple knee ligaments reconstructed. Descriptive statistics of athletes' demographics are presented in Table 1.

Table 1. Summary of participant information

Participant Information (N = 35)	Values[a]
Months post-ACLR	8.9 (2.5)
Body mass, kg	78.2 (16.9)
Female, n (% of total participants)	20 (57%)
Graft type, n (% of total participants)	
Bone-patellar tendon-bone	29 (82.9%)
Hamstring tendon	4 (11.4%)
Allograft	2 (5.7%)
Biomechanical measures	
Maximal comfortable running speed, m/s	4.0 (0.5)
Jump height, cm	31.9 (8.1)
$PKEM_{RUN}$	
Surgical, N·m/kg	1.59 (0.48)
Nonsurgical, N·m/kg	2.36 (0.39)
LSI (%)	67.6 (18.8)
CMJ $PKEM_{TO}$	
Surgical, N·m/kg	1.14 (0.24)
Nonsurgical, N·m/kg	1.58 (0.28)
LSI (%)	83.7 (10.6)
CMJ $PKEM_{LAND}$	
Surgical, N·m/kg	1.16 (0.33)
Nonsurgical, N·m/kg	1.73 (0.43)
LSI, %	80.2 (16.6)
Patient-reported outcomes	
IKDC	81.5 (12.1)
Knee extension peak torque measures	
Isometric	
Surgical, N·m/kg	2.62 (0.76)
Nonsurgical, N·m/kg	3.46 (0.79)
LSI, %	76.3 (16.9)
Isokinetic	
Surgical, N·m/kg	1.99 (0.54)
Nonsurgical, N·m/kg	2.60 (0.44)
LSI, %	77.3 (18.9)

ACLR, anterior cruciate ligament reconstruction; CMJ, countermovement jump; IKDC, International Knee Documentation Committee Subjective Knee Form score; LAND, landing phase: LSI, limb symmetry index; PKEM, peak knee extensor moment; RUN, stance phase of running; TO, takeoff phase.
[a]Values are presented as mean (SD) unless otherwise indicated.

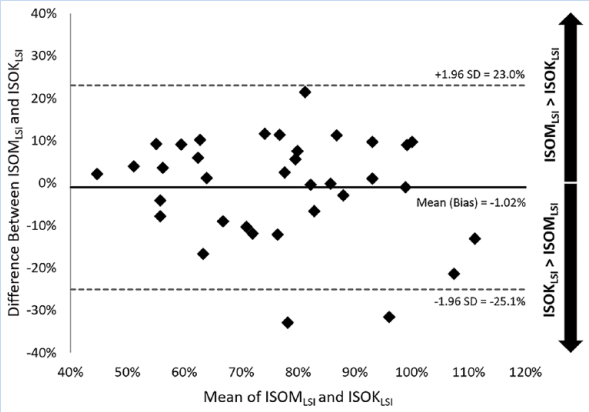

Figure 1. Bland-Altman plot for difference in quadriceps strength symmetry quantified via isometric assessment and isokinetic assessment. Positive *y*-axis values indicate $ISOM_{LSI} > ISOK_{LSI}$, while negative *y*-axis values mean $ISOK_{LSI} > ISOM_{LSI}$. $ISOK_{LSI}$, isokinetic knee extension torque LSI; $ISOM_{LSI}$, isometric knee extension torque LSI; LSI, limb symmetry index.

Quadriceps Strength, Functional Knee Kinetics, and Patient-Reported Outcomes

$ISOM_{LSI}$ was 76.3 ± 16.3% and $ISOK_{LSI}$ was 77.3 ± 18.9%. The average coefficient of variance across the 5 isokinetic trials was 7.8% (nonsurgical) and 8.3% (surgical). The maximally comfortable running speed utilized for analyses in this study was 4.0 ± 0.5 m/s (6:37 min/mile). $PKEM_{RUN}$ LSI was 67.6 ± 18.8% (surgical limb, 1.6 ± 0.5 N·m/kg; nonsurgical limb, 2.4 ± 0.4 N·m/kg). $PKEM_{TO}$ LSI was 83.7 ± 10.6% (surgical limb, 1.1 ± 0.2 N·m/kg; nonsurgical limb, 1.6 ± 0.3 N·m/kg). $PKEM_{LAND}$ LSI was 80.2 ± 16.6% (surgical limb, 1.2 ± 0.3 N·m/kg; nonsurgical limb, 1.7 ± 0.4 N·m/kg). Mean IKDC score was 81.5 ± 12.1. Descriptive statistics of the athletes' biomechanics, strength, and patient-reported outcome data are shown in Table 1.

Comparison of Isometric and Isokinetic Quadriceps Strength Assessment

The Bland-Altman plot of isometric versus isokinetic quad strength LSI is depicted in Figure 1. Mean difference was 1.0%, with a 95% limits of agreement of -25.1% to 23.0%. RMSE between LSIs was 12.1% and ICC was 0.770.

Relationships Between Quadriceps Strength, Functional Knee Kinetics, and Patient-Reported Outcomes

Moderate-to-strong associations between $ISOM_{LSI}$ and $PKEM_{RUN}$ LSI, $PKEM_{TO}$ LSI, and $PKEM_{LAND}$ LSI were observed. Moderate associations were observed between $ISOK_{LSI}$ and $PKEM_{RUN}$ LSI, $PKEM_{TO}$ LSI, and $PKEM_{LAND}$ LSI. A weak-to-moderate association between IKDC scores and $ISOM_{LSI}$ was observed. A moderate association between $ISOK_{LSI}$ and IKDC scores was observed (Table 2 and Figure 2).

DISCUSSION

To advance understanding and clinical application of quadriceps strength assessments in post-ACLR athletes, this study aimed to directly compare isometric and isokinetic knee extension efforts. Consistent with our first hypothesis, isokinetic and isometric assessments of knee extension torque demonstrated strong agreement, supporting the use of either contraction type for clinical assessment of quadriceps strength. However, the wide 95% limits of agreement (±24.1%) between contraction types are indicative of significant individual variation and support the use of a consistent method of assessment throughout the rehabilitation process. This limits of agreement is comparable with previously reported minimally detectable changes of 15% and 25% for isometric (at 90° of knee flexion) and isokinetic (at 60 deg/s) knee extension peak torque, respectively, in people with knee osteoarthritis.[22] Contrary to our secondary hypotheses, isometric strength LSI was moderately-to-strongly associated with PKEM LSIs during running and jumping, while isokinetic LSI was moderately associated with PKEM LSIs. Further, isometric LSI was weakly-to-moderately associated with IKDC scores, while isokinetic LSI was moderately associated with IKDC scores. Knee extension torque assessment is an integral component in the return-to-sport decision-making process. As abnormal movement biomechanics have been associated with ACL injury,[42] and may contribute to posttraumatic knee osteoarthritis,[10] the findings from this study further support the use of knee extension torque assessment as part of the return-to-sport testing battery regardless of the contraction type.

Table 2. Correlations between quadriceps strength symmetries, functional knee kinetics, and patient-reported outcomes

Knee Extension Peak Torque LSI	$PKEM_{RUN}$ LSI (*r*, [95% CI] *P* value)	$PKEM_{TO}$ LSI (*r*, [95% CI] *P* value)	$PKEM_{LAND}$ LSI (*r*, [95% CI], *P* value)	IKDC (*r*, [95% CI], *P* value)
Isometric	0.63, [0.37,0.80] <0.01	0.80, [0.64,0.90] <0.01	0.64, [0.39,0.80] <0.01	0.39, [0.06,0.64] 0.02
Isokinetic	0.41, [0.09, 0.65] 0.01	0.58, [0.31, 0.77] <0.01	0.58, [0.31,0.77] <0.01	0.49, [0.19,0.71] <0.01

Correlations are reported as Pearson's *r*. CMJ, countermovement jump; IKDC, International Knee Documentation Committee Subjective Knee Form score; LAND, landing phase of CMJ; LSI, limb symmetry index; PKEM, peak knee extensor moment; RUN, stance phase of running; TO, takeoff phase of CMJ.

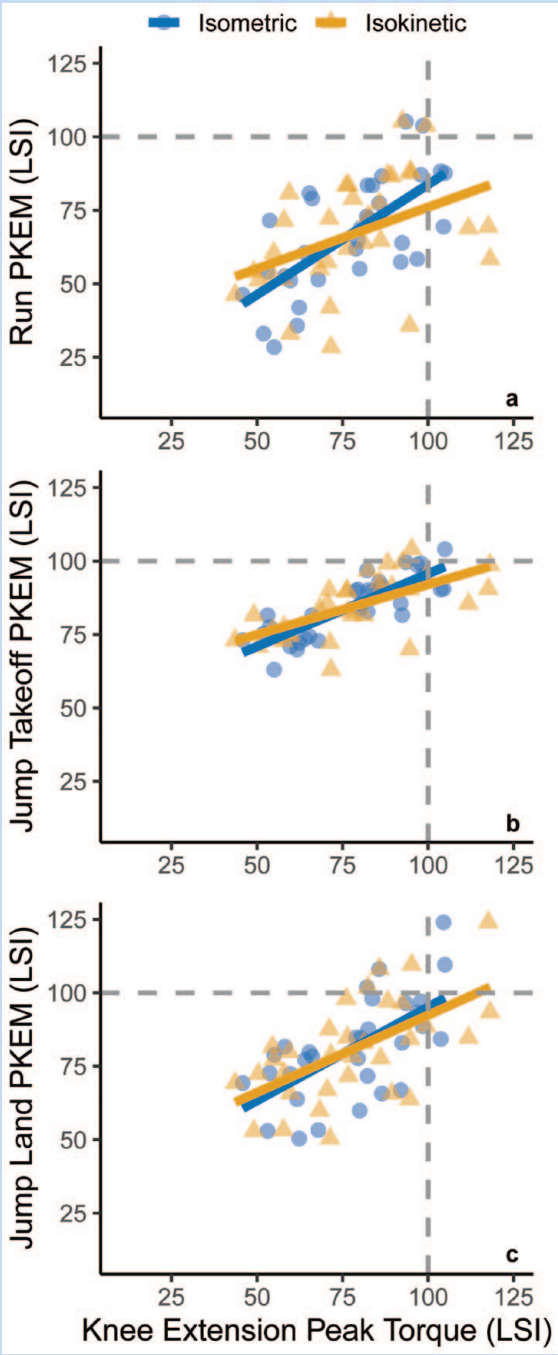

Figure 2. Scatterplot and simple linear regression for both isokinetic (yellow) and isometric (blue) knee extension strength symmetry (a) to PKEM symmetry during running, (b) concentric phase of the CMJ and (c) landing phase of the CMJ. CMJ, countermovement jump; LSI, limb symmetry index; PKEM, peak knee extensor moment.

Our results of a strong agreement between isometric and isokinetic LSI with large variation on the individual level are remarkably similar to those of Reinking et al,[50] who reported a Pearson's *r* of 0.76 for the correlation of isometric and isokinetic quadriceps strength symmetry in 23 patients with unilateral knee dysfunction. Similarly, Knezevic and colleagues[23] reported Pearson's *r* values ranging from 0.66 to 0.81 for isometric and isokinetic quadriceps LSI comparisons in 20 male subjects before and up to 6 months after ACLR. Our observed agreement between isokinetic and isometric quadriceps strength symmetry in collegiate athletes 9 months post-ACLR is also similar to what has been reported previously in patients with knee or hip replacement surgery,[27] in that both modes of contraction were valid in identifying quadriceps muscle weakness (though there were some discrepancies in the magnitude of asymmetry).

Both isometrically and isokinetically quantified quadriceps strength have been shown to be associated with movement mechanics and patient-reported outcomes post-ACLR.[18,26,30,59] The current study is the first to compare both isometric and isokinetic testing methods on collegiate athletes after ACLR at a clinically relevant timepoint. The results of this study add to the current body of literature, demonstrating that isometric quadriceps strength symmetry is moderately-to-strongly associated with running and jumping knee kinetics and weakly-to-moderately associated with patient-reported outcomes. Conversely, isokinetic strength symmetry was moderately associated with both patient-reported outcomes and functional knee kinetics. Similar to our findings, Hunnicutt et al[18] reported a significant correlation (Spearman's rank correlation coefficient, *r* value of 0.38) between isokinetic quadriceps strength symmetry and IKDC scores in 30 patients 6 to 23 months post-ACLR, but no association with isometric assessment. Hunnicutt et al[18] also observed moderate associations (Spearman's *r* values of 0.54-0.57) between isokinetic and isometric quadriceps strength symmetry and single-leg hop symmetry. Lienhard et al[29] also reported moderate to strong correlations (Pearson's *r* value of 0.62-0.73) between isometric and isokinetic assessments of involved limb quadriceps strength, gait characteristics, and patient-reported outcomes in 29 patients 6 months after total knee arthroplasty. Ultimately, quadriceps strength symmetry is important to functional knee kinetic symmetry and subjective outcomes in patients post-knee surgery but the strength of the association may vary depending on mode of contraction and variables of interest.

Running is an integral component of most sports and is commonly resumed around 3 months postoperatively, often without objective criteria to support this clinical decision.[49] At 3 months post-ACLR, quadriceps strength deficits are nearly universal[45]; however, there are few published investigations of the relationship between quadriceps strength and running biomechanics.[40] Current evidence suggests there are pronounced abnormalities in surgical knee kinematics and kinetics during running, which persist beyond the typical return-to-sport timeframe.[24,37] In the current study, isometric peak torque LSI accounted for 39.7% of the variance ($r = 0.63$ [95% CI, 0.37-0.80]) in $PKEM_{RUN}$ LSI, while isokinetic peak torque LSI accounted for 16.8% of the variance ($r = 0.41$ [95%

CI, 0.09-0.65]) in $PKEM_{RUN}$ LSI. Quadriceps strength symmetry, particularly isometric symmetry, may provide valuable insight into running biomechanics symmetry, which is clinically meaningful as detailed assessment of running mechanics is primarily limited to research settings.

The CMJ is a common test of athletic performance,[43] and jumping and landing tasks are a recommended component of return-to-sport testing.[32] Quadriceps strength symmetry has been previously associated with landing biomechanics,[20,30,39,52] as well as performance symmetry.[9,18,26,47,61] We observed significant associations between both isometric and isokinetic knee extension torque symmetry and CMJ performance. Isometric peak torque LSI accounted for 40.9% of the variance (r = 0.64 [95% CI, 0.39-0.80]) in $PKEM_{LAND}$ LSI, whereas isokinetic peak torque LSI accounted for 33.6% of the variance (r = 0.58 [95% CI, 0.31-0.77]) in $PKEM_{LAND}$ LSI. The concentric phase of the CMJ is of interest, particularly in collegiate athlete populations, as it is commonly used as an assessment of athletic performance.[4] Interestingly, knee extension torque symmetry was associated most strongly with $PKEM_{TO}$, accounting for 64.0% (r = 0.80 [95% CI, 0.64-0.90]) and 33.6% (r = 0.58 [95% CI, 0.31-0.77]) of the variance in $PKEM_{TO}$ symmetry for isometric and isokinetic tests, respectively. Knee kinetics during the concentric phase of the countermovement jump have been reported previously as more strongly associated with knee extension strength than knee kinetics of the landing phase.[35] Our results further support quadriceps strength assessment as a key component of ACLR rehabilitation efforts in athletes, as quadriceps strength symmetry is important to achieving knee kinetic symmetry during the concentric and landing phases of the CMJ.

It is often assumed that isokinetic assessment of quadriceps muscle strength may be more clinically meaningful than static muscle assessments, as athletic tasks are largely dynamic. However, isokinetic assessment occurs under conditions of constant velocity, which is also not representative of contractile mechanics in performance of athletic movements. Further, in a similar population of collegiate athletes, angle-specific analyses of isokinetic knee torques demonstrated the greatest magnitude deficits in deeper knee flexion angles, similar to the isometric testing position utilized in this study.[25] Our results suggest that while isokinetic quadriceps strength symmetry is meaningful to functional knee kinetics post-ACLR, isometric strength symmetry may be a better indicator of functional knee kinetic symmetry, particularly in regard to the stance phase of running and the takeoff phase of the CMJ.

The IKDC is a commonly employed patient-reported outcome to assess self-perceived function after ACLR.[19] IKDC scores have been shown to be meaningfully associated with movement biomechanics and reinjury after ACLR.[12,38] Our results demonstrate that both isometric (r = 0.39 [95% CI, 0.06-0.64]) and isokinetic (r = 0.49 [95% CI, 0.19-0.71]) peak knee extension torque LSIs are weakly-to-moderately associated with IKDC scores, largely consistent with previous literature.[63]

It is unclear why we observed such a large individual variation in quadriceps strength symmetry between isometric and isokinetic efforts. In 4 of 35 athletes, the LSI difference between modes was >20%, though the majority of athletes (22 of 35) recorded $ISOM_{LSI}$ and $ISOK_{LSI}$ differences of <10%. It is possible that efforts could be pain-limited, as anterior knee pain is commonly reported post-ACLR,[51] particularly after bone-patellar tendon-bone autografts,[62] and may be joint-angle dependent. Although we did not collect numerical pain ratings, all athletes in this study were able to complete the testing protocol without verbalizing or demonstrating significant pain beyond what they experienced during their typical rehabilitation exercises. All athletes had previously performed at least 1 postoperative session of both contraction types, so it is unlikely that the observed discrepancies were due to learning effect. The most likely explanation appears to be a difference in voluntary effort between contraction types, but this is speculative and cannot be confirmed or quantified. Strong and consistent verbal encouragement was provided for all voluntary strength assessments in attempt to properly motivate subjects to produce maximal knee extension torque, but since quadriceps strength testing was just one of a battery of tests, the influence of mental or physical fatigue may have contributed to the larger individual variations.

LIMITATIONS

Due to the relatively small sample size and narrow population, this study should be considered preliminary and a starting point for future research. All testing in the current study was performed with an isokinetic dynamometer. As such, the study results cannot be generalized to other methods of strength assessment, such as handheld dynamometers for isometric strength assessments. We did not utilize the isokinetic peak torque coefficient of variance in real time to determine whether isokinetic efforts were acceptable. However, the recorded average coefficient of variance across the group was 7.8% (nonsurgical) and 8.3% (surgical), well within the range of recommend values (<12% to 15%).[5] The majority of athletes in this study underwent ACLR with bone-patellar tendon-bone autografts. Recent investigations show disparities in quadriceps strength by graft type.[54,58] However, we would not expect mode of contraction to influence strength symmetry by graft type, as these graft-specific strength profiles have been documented for both isometric and isokinetic assessments.[54,58]

The sample for this study consists of Division I collegiate athletes at a clinically relevant time point with a small variation in time from surgery (8.9 ± 2.5 months). As such, all subjects underwent similar rehabilitation and return-to-sport testing. This strengthens the internal validity of our study, but may limit the generalizability of the results to other populations. As the data for this study came from a sample of convenience, we were not adequately powered to assess for confounding variables, such as sex, sport, graft type, or concomitant surgical procedure.

CONCLUSION

Isokinetic and isometric assessments of knee extension torque symmetry in collegiate athletes 9 months post-ACLR demonstrated strong agreement, supporting either contraction

type for the assessment of quadriceps strength. Significant individual variation suggests mode of contraction should be consistent throughout the postoperative assessment period. Moderate-to-strong associations were observed between isometric quadriceps strength LSI and functional knee kinetics, while moderate associations were observed between isokinetic strength LSI and functional knee kinetics. IKDC scores were weakly-to-moderately associated with isometric and moderately associated with isokinetic peak knee extension torque symmetry.

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